Further paleomagnetic results from the Seal Group igneous rocks, Quebec[1]

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This paleomagnetic study involves further demagnetization (alternating field, thermal) of specimens from part of the previously published Seal Group (volcanics and sills). A previous study uncovered two stable magnetizations nominated M2 (herein *B*) and M3 (herein *A*). In the present work, the *A* magnetization has been better defined, found to contain reversals of direction, and has been found also in the sills (6 sites, specimen unit weight; 289°, +11°; α_{95} = 11°) as well as the volcanics (5 sites; 293°, +13°; α_{95} = 13°) for a combined direction with respect to bedding of 291°, +12° (11 sites, R = 10.76, α_{95} = 7°). Of the total number of specimen directions (63; 1 or 2 per sample), 17 were obtained by vector subtraction (VS); all but four of the latter were the means of at least two determinations. It is suggested that the *A* magnetization residing in hematite was acquired first, and *B*, residing in magnetite, later, as a result of remagnetization during the Grenville 'Orogeny'. At several sites, where there is no evidence of hematite, a reverse *B* direction may be present. The *A* pole is at 17°N, 195°E (δp = 4°, δm = 7°), near the Seal Group red bed pole (05°N, 205°E), and the *B* pole (direction with respect to horizontal) at 17°S, 158°E (7°, 9°).

It is apparent in this and other studies that specimens that contain both normal and reverse magnetizations display a very characteristic behaviour under demagnetization. This behaviour and its implications for deriving good approximations to the true direction by the use of VS—even when remanent coercive force or unblocking temperature spectra overlap substantially—are revealed by means of a model using artificial data.

On décrit l'aboutissement d'une étude plus approfondie découlant de traitements additionnels (thermiques et par champs alternatifs allant jusqu'à 290 mT) pratiqués sur une partie des spécimens du groupe du Seal (les volcaniques et filon-couches) dont certains résultats ont déjà été publiés. Une étude antérieure révéla l'existence de deux aimantations appelées M2 (ici *B*) et M3 (ici *A*). Dans le travail présent la direction de l'aimantation *A* est mieux définie et souvent recouvrée dans les deux sens. De plus, elle est observée aussi bien dans les filoncouches (6 sites; 289°, +11°; α_{95} = 11°, l'unité de poids étant donnée au spécimen) que dans les volcaniques (5 sites, 293°, +13°; α_{95} = 13°) pour nous donner une direction moyenne de 291°, 12° (11 sites; R = 10.76; α_{95} = 7°) après correction pour le pendage. Du nombre total des déterminations de direction (63; 1 ou 2 par échantillon), 17 furent obtenues par soustraction de vecteur (VS); chacune, excepté quatre du dernier groupe, étant la moyenne d'au moins deux observations indépendantes. Notre interprétation est que l'aimantation *A* qui est portée par l'hématite fut acquise en premier et que *B* portée par la magnétite est l'effet d'une ré-aimantation qui eut lieu plus tard, probablement lors de l'orogénèse du Grenville. A plusieurs sites, où l'hématite semble absente, une direction *B* à polarité négative (renversée) parait être présente. Le pôle *A* se situe à 17°N, 195°E (δp = 4°, δm = 7°) près du pôle donné par les couches rouges du groupe du Seal (05°N, 205°E), et le pôle *B* (dérivé de la direction par rapport à l'horizontale) à 17°S, 158°E (7°, 9°).

Il appert, d'après cette étude ainsi que d'autres publiées tout récemment, que des spécimens contenant des remanences dont la direction est dans les deux sens manifestent un comportement très caractéristique pendant les traitements de désaimentation. Ce comportement et ses implications en ce qui a trait aux approximations de la vraie direction que l'on peut obtenir par analyse vectorielle—même quand les spectres de coercitivité et de température de dégagement chevauchent sensiblement—sont exposés à l'aide d'une modèle employant des données artificielles.

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Introduction

In paleomagnetic studies it is not uncommon to find that only one magnetic direction has been derived from a rock unit. This is usually the case for basic igneous rocks—the prime target for paleomagnetic studies—from the Phanerozoic Eon. When one begins analyzing rocks from the Precambrian, it is apparent that even igneous rocks (e.g., the Nipissing diabase, Roy and Lapointe 1976) may have acquired several magnetizations, each with a different direction. The questions then arise: What is the sequence in which the magnetizations were acquired? and what age is associated with each? The answer to the former question can sometimes be determined unequivocally through field tests or analysis, but usually turns out to be a

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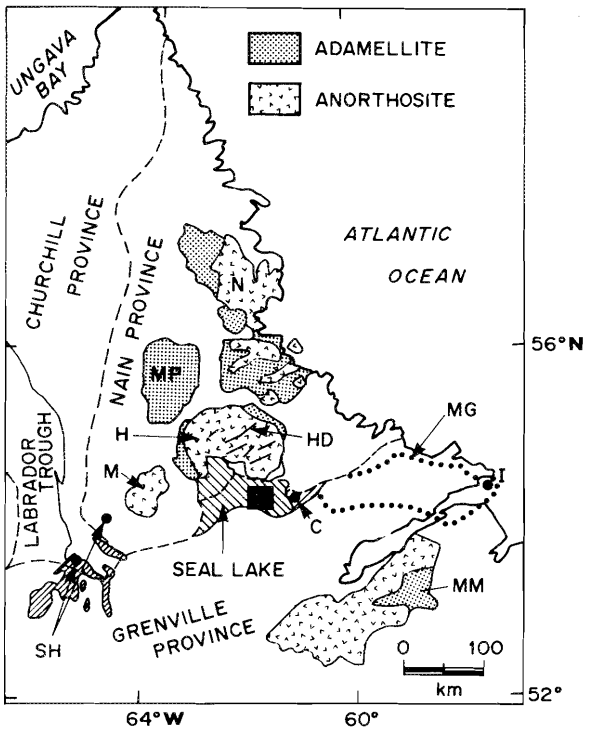


FIG. 1. Location map of the Seal Group study (black square) and other previous pertinent studies (identified by letters) in the Nain and northeastern Grenville Structural Provinces. Letters refer to upper Croteau Group volcanics (C; Roy and Fahrig 1973), Harp Lake Complex (H; Irving *et al.* 1977), Harp dikes (HD; Irving *et al.* 1977), Indian Harbour dikes (I; Murthy and Deutsch 1971), Michikamau anorthosite (M; Murthy *et al.* 1968; Emslie *et al.* 1976), Michael gabbro (MG; Fahrig and Larochelle 1972), Mealy Mountain anorthosite suite (MM; Fahrig *et al.* 1974), Mistastin pluton (MP; Fahrig and Jones 1976), Nain anorthosite (N; Murthy 1978), and Shabogamo gabbro (SH; Fahrig *et al.* 1974).

matter of judgement. The answer to the latter is still problematical, and again becomes a reasoned judgement.

These questions are basic to all paleomagnetic studies and indicate the crux of the present work, which involves a re-interpretation of directions derived from previous work on the Seal Group (Roy and Fahrig 1973, henceforth in this paper called RaF) in the light of additional data. The new data have come from the further treatment of a number of previously used specimens.

The Seal Group essentially comprises volcanics, diabase sills, and sediments, which lie in a belt along the northeastern Grenville Front (Figs. 1 and 2). To the east are the upper Croteau Group volcanics, which predate the Seal Group (Marten and Smyth 1975). Field data suggest that Seal Group rocks lie unconformably on the Harp Lake intrusion exposed to the north of the Grenville Front, though an actual contact has not been observed (Irving *et al.* 1977; Meyers and Emslie 1977). In the area of consideration the Seal Group has been folded during a possibly complex deformational sequence, but aside from possible hydrothermal alteration, there is no regional metamorphic overprint present. The Seal Group has been dated at 1350 ± 92 Ma (Rb/Sr, $\lambda = 1.39 \times 10^{-11}$ yr^{-1}, W. R. A. Baragar *in* Wanless and Loveridge 1978) and the Harp Lake complex at about 1450 Ma (Rb/Sr, Krogh and Davis 1973). Seal Group rocks may be contemporaneous with other rock units in the Nain and Grenville Provinces: namely, the Harp dikes (K/Ar, 1316 ± 94 Ma), which intrude the Harp Lake Complex, the Michael gabbro (Rb/Sr, 1488 ± 108 Ma) farther to the east, and the dikes that intrude the Mealy Mountains anorthosite (Meyers and Emslie 1977; Emslie 1978). Seal Group sites mentioned in the text are shown in Fig. 1.

In the previous paleomagnetic study (RaF) two directions were obtained from the Seal Group rocks and one from the adjacent Croteau Group volcanics. Seal Group volcanics yielded directions M2 (146°, +43°; herein nominated *B*) and M3 (292°, +13°; herein, *A*), and Seal Group diabases just *B* (143°, +37°): all with respect to bedding. The *A* direction represented by only 6 samples from 3 sites appeared to be related to hematization, which according to RaF "clearly postdates the consolidation of the flows as its presence is controlled by columnar joints, irregular fractures, or breccia zones" (p. 1292). Red sandstones from the Seal Group at the top of the section yielded only an *A*-type direction (276°, +05°: called the Group 1 direction in RaF). RaF considered the *A* direction of the igneous rocks and the *A*-type direction of the sandstones to have been acquired later than the *B* direction, since the hematization of these rocks was a later event. In addition the *B* direction is close to that obtained from the Michael gabbro (138°, +31°) to the northeast, which shows no evidence of post-crystallization alteration or the *A* direction. Both the Seal Group and the Michael gabbro lie along the Grenville Front, and it was thought both by Fahrig and Larochelle (1972) and also by RaF that the *B* direction was the result of the tectonic rotation of a single block about a vertical axis. RaF estimated this rotation to have been 55° using the red bed results.

Methods

Recognition of Multicomponent Directions

There are a number of anomalous directions in the Seal Group which complicate the picture of two simple directions, *A* and *B* (see e.g. Fig. 4). Normally one would either average these with a bonafide direction, nominate them as entirely new directions, or reject them on the basis of some criteria. The procedure would depend on the evi-

dence. In the present case all three procedures have been used to some extent based on the following simple model: namely, that the directions that appear to be anomalous are composed of (predominantly) normal and reverse *A* directions.

This model is based on two lines of evidence. First, several independent studies have confirmed that anomalous directions can be the vector resultants of two or more unresolved magnetizations (e.g., Roy and Robertson 1968; Roy and Park 1974; Buchan and Dunlop 1976; Irving *et al.* 1977). And second, under treatment several specimen directions migrate to the reverse *A* (easterly) direction (e.g., Fig. 6).

This model is favoured by another line of evidence—a conclusion drawn from this and other studies—that specimens containing opposed magnetizations often tend to display some sort of erratic and (or) inhomogeneous behaviour, which shows up on the stereographic plots, demagnetization curves, vector diagrams, and measurement sheets. Roy and Park (1974) demonstrated that non-uniformity of component measurements was a sign of the presence of opposed magnetizations. Larson and Walker (1975) and Roy (1977) have shown that different parts of red bed samples carry magnetizations of opposite polarity. Such antiparallel magnetizations were also proposed by Roy and Park (1969) and by Irving *et al.* (1977) to account for some intermediate directions. Recently, Murthy (1978) has postulated the presence of opposed magnetizations to explain the erratic behaviour of specimens undergoing AF demagnetization. Some of these characteristics are implicit in a theoretical model given in the Addendum. Here it is shown that depending on the ambient conditions and the degree of overlap of the remanent coercive force (rcf) and (or) unblocking temperature (T_{UB})[2] spectra one or the other component predominates. The resulting manifestation is the type of erratic behaviour ubiquitous in the present study.

Analysis of Multicomponent Directions

In studies of apparently anomalous directions found in red sediments Roy and Park (1974) showed that most of these were a manifestation of vector resultants of magnetizations associated with at least three different phases of hematite deposited and (or) developed over a period during which the earth's field reversed. Study of these phases in these and similar red sediments was greatly facilitated by the use of acid leaching to remove or isolate the magnetic phases. Prior to the time when acid leaching was extensively used, RaF used an empirical formula based on non-uniformity in the measurements obtained from an astatic magnetometer to at least reject unresolved multicomponent magnetizations. The method of acid leaching is not practical with igneous rocks—as in the present study—owing to the largely impervious nature of these rocks. And, an empirical formula to reject directions was not easily adapted to measurements carried out on the Schonstedt spinner magnetometer (DSM). One method that can be used for igneous (and other) results is vector subtraction (VS).

When either the rcf or T_{UB} spectra of the various components occupy different ranges, they can be experimentally separated. When spectra coincide no resolution is possible by experimental or vectorial methods. In a third case where spectra overlap (sometimes substantially) the separation can often be effected by the method of VS. In the present study there are many apparently anomalous directions which behave in a manner characteristic of multicomponent magnetizations having overlapping rcf and T_{UB} spectra, and many of these have been resolved by the method of VS. The method is justified by reference to the theoretical model given in the Addendum. It is emphasized that the method described applies only to magnetizations that are nearly reversed.

Specimen and site-level analyses are emphasized in this paper, but sample analysis has also been carried out and averages in tables quoted for comparison. The specimen level is used, because often specimens containing more than one remanence behave in a very individual manner upon treatment. A sample with one of its component specimens yielding good data might under ordinary circumstances be eliminated owing to a large discrepancy of direction between specimens. Working at the specimen level ensures that only data from truly hopeless specimens are eliminated from consideration. Specimen directions quoted in Table 1 consist of: (1) end-points (E), (2) means of several directions close to E (M), (3) vector subtraction directions (V), and (4) the mean of several V, or mixed V and M directions (VM). Case (2) is used because in several instances where both normal

[2]The origin of a magnetization is usually difficult to establish. Although thermal experiments may be useful in identifying some of the magnetic carriers, they do not necessarily provide one with information about the origin of the magnetization (whether a DRM (depositional), CRM (chemical), TRM (thermal), or T-CRM (thermo-chemical remanent magnetization)) or the temperature at which it was acquired. Hematite for example could be formed by oxidation, hematization, or metasomatism at temperatures as high as the Curie temperature (~ 680°C), as low as 200°C for igneous rocks, or even lower for sedimentary rocks; yet not be demagnetized in laboratory experiments until temperatures of 600 to 680°C are reached. In view of this it appears preferable to make a clear distinction between the temperature at which the remanence is unblocked (T_{UB}) and the blocking temperature at which it was presumably acquired (T_B)

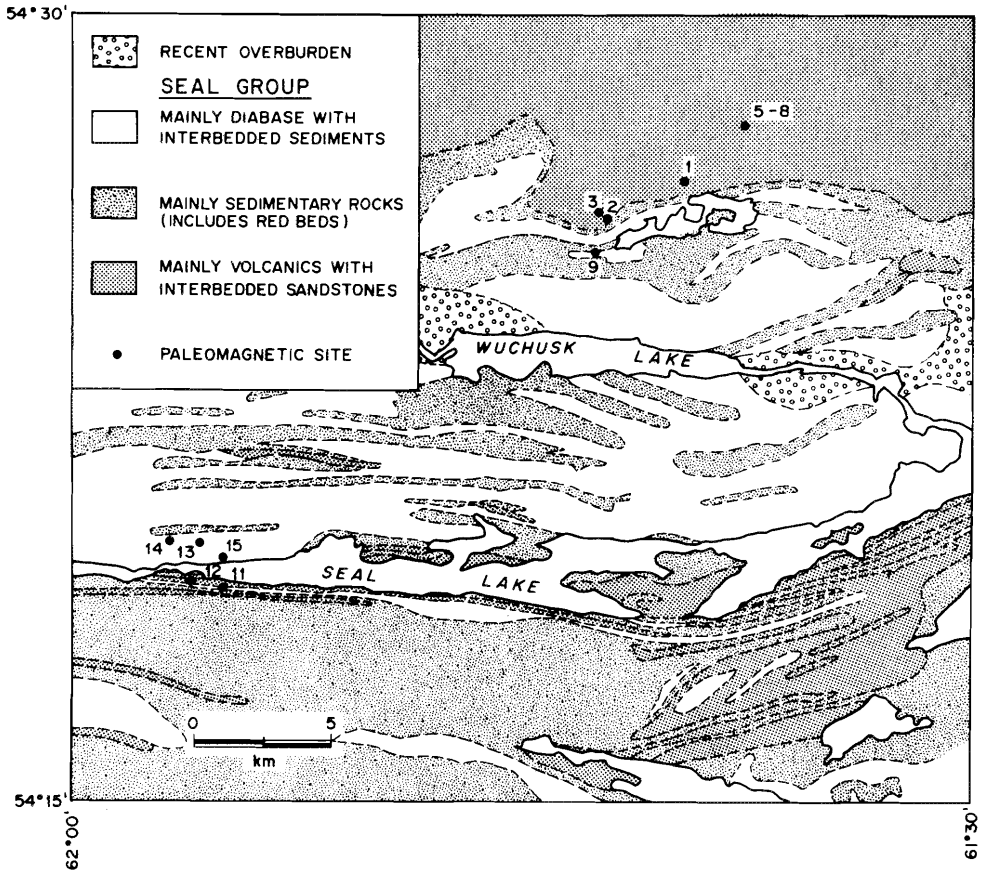


FIG. 2. Site locations of Roy and Fahrig (1973) in area represented by black square in Fig. 1.

and reverse components are present and one substantially dominates the other, no true end-point occurs on demagnetization, only an erratic cluster of directions. Case (3) is dependent on a knowledge of the approximate true end-point direction and has been employed judiciously according to procedures outlined in the Addendum. In (4) several normal and reverse directions may have been obtained. Therefore they have been averaged separately, given unit weight, and then meaned together.

In the previous work of RaF only one or two pilot specimens from each igneous site had been cleaned in detail with alternating fields (AF) up to 290 mT, and no thermal work had been attempted. Remaining specimens were mainly demagnetized in the low fields (10–30 mT) indicated by the pilot specimens, thus yielding the *B* direction. In the present work most of these specimens were either further demagnetized in higher fields or thermally demagnetized; the method of cleaning being determined by the behaviour of pilot specimens. Since specimen behaviour was often complex, treatment steps were generally more numerous, with smaller increments between them (5 mT intervals below 100 mT; 2 to 5°C intervals near the Curie point of hematite).

The instrumental set-up for determining the directions of magnetization has been indicated by RaF, with the exception that all present measurements have been made on a DSM.

Results

The predominant direction uncovered in this study by the methods above—the *A* direction—is detailed in Table 1, summarized in Table 2, and illustrated in Fig. 3. This direction (Table 2, entry 9: 291°, +12°; α_{95} = 7°) differs little from the previously determined *A* (M3) direction (292°, +13°) of RaF obtained from six extremely stable specimens which are also included in the new determination. Of the sites available for further study (1–3, 5–9, 11–15) only 5 and 12 did not yield *A* directions. At site 5 exploratory treatment produced erratic directions and indicated that further detailed work would likely not prove fruitful. On the other hand previous work on site 12 indicated that it might be possible to isolate an *A* direction with the aid of thermal treatment, since fields up to 290 mT moved the initial directions very little. Although the at-

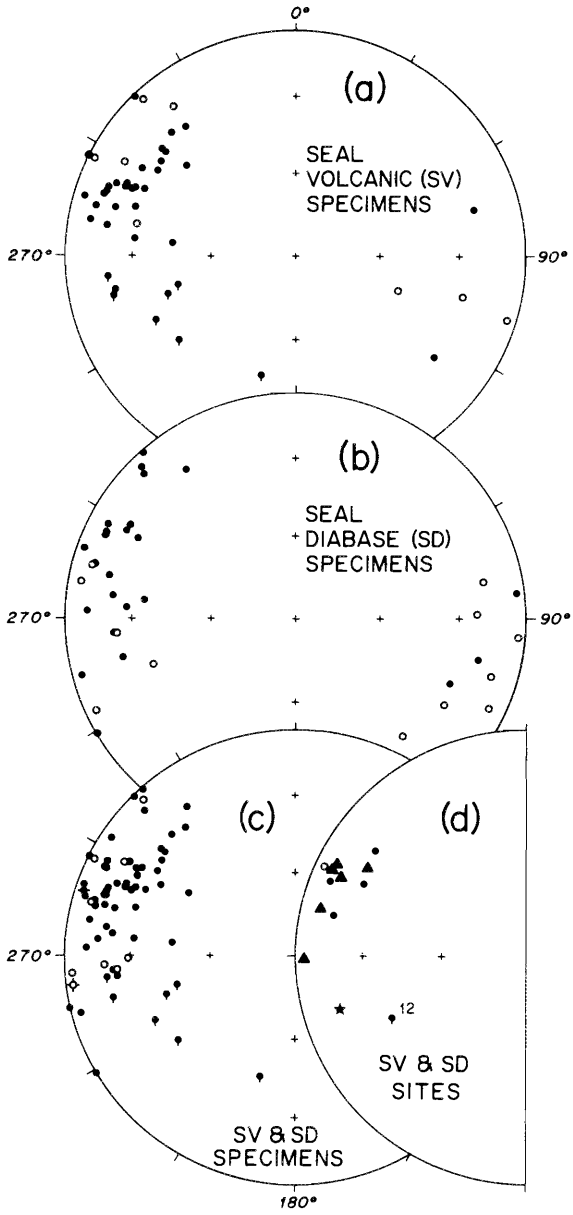


FIG. 3. The A direction as depicted by AF and thermally demagnetized specimens (a, b, c) and sites (d). In (a), (b), and (c) dots (circles) represent positive (negative) inclinations. The dots and circles in (d) are volcanics; the triangles, diabases. In (c) easterly directions have been reversed, and either plotted as is (crossed symbols) or averaged with existing westerly directions from the same specimen. Site 12 specimens and mean are indicated by 'tailed' dots. Directions in this and succeeding figures are given with respect to the bedding plane. Equal area projection.

tempt was unsuccessful, there is certainly a tendency toward A (Fig. 4); therefore the site has been listed but excluded from the final results.

Both thermal and AF treatments have been used in isolating the A direction. A good example is shown in Fig. 5, where one specimen of a sample has been demagnetized by AF and the other by a two-stage process—AF followed by thermal. There are several examples like this where thermal and AF results are in excellent agreement (see e.g. site 1, 5A and 5B, Table 1), and this agreement is emphasized in a comparison between the averages of all thermal results (18 directions) with all AF results (45 directions) (Table 3).

Site 15 presents good examples of A directions found both directly (Fig. 6) and by VS (Figs. 7 and 8). The A directions found directly (Fig. 6) are only stable up to about 550°C. Both easterly and westerly A magnetizations are recovered by VS involving specimen 6C (Fig. 7). This is a spectacular example of what must be the superimposition of two rcf spectra (E and W components) whose intensity ratio is nearly unity, but where, first one component, and then the other, dominates. Another example shows the recovery of the westerly component by VS of an apparent end-point from all previous values (Fig. 8).

A second direction was uncovered by further AF work on specimens of site 2 (Fig. 9). This direction is rather erratic, but could indicate a reversal to B. The error circle of the B direction ($\alpha_{95} = 6°$, from average of directions at 20 mT) when reversed does not overlap that calculated around the mean of the final cleaned directions ($\alpha_{95} = 14°$, 35–85 mT, Fig. 9). It was first thought that these latter directions were merely the resultants of normal and reverse A directions; however, this is one of the three sites (2, 9, 15) for which no hematite was detected in the

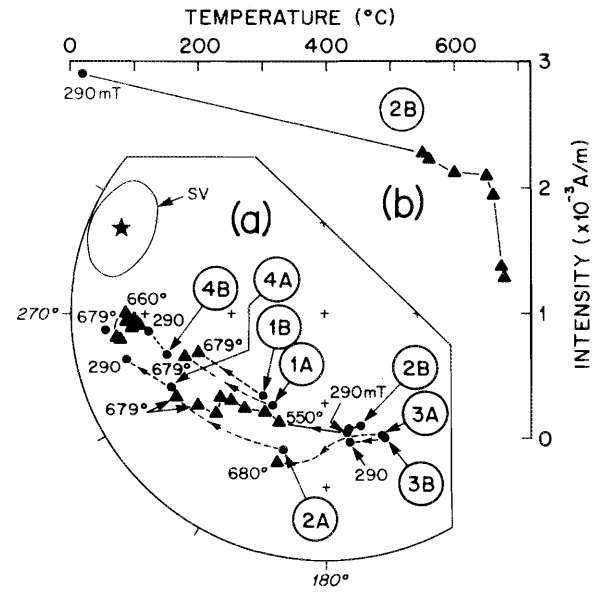


FIG. 4. (a) AF and thermal demagnetization of specimens from site 12. (b) Thermal decay curve of one specimen. SV is the mean A direction (with error circle) for the Seal Group volcanics.

TABLE 1. *A* direction results

Specimen	Direction $D°, I°$ (R)	VS ($D°, I°$)	w, e	Type	Trt °C/mT
		Seal volcanics			
Site 1	(54.5°N, 61.6°W)				
1A	320, +27		1,0	E	75
1B	301, +45 (1.97)		1,1	M	652°
2A	315, +25 (3.70)		4,0	M	70
2B	306, +29 (5.90)		6,0	M	70
3A	276, +30 (4.91)		5,0	M	70
3B	292, +24 (1.99)		2,0	M	70
4A	280, +10 (4.82)		5,0	M	70
4B	279, +18 (4.87)		5,0	M	70
5A	289, +14 (1.89)		2,0	M	600°
5B	288, +13 (6.96)		7,0	M	70
6A	292, +21 (3.85)		4,0	M	70
6B	293, +25 (5.62)		6,0	M	70
7A	290, +14 (1.99)		2,0	M	75
7B	309, +26 (6.98)		7,0	M	70
8A	294, +29 (6.91)		7,0	M	70
8B	293, +25 (6.98)		7,0	M	70
16	**294, +24 (15.54, 33, 7)**		**24°N, 164°W**		
Site 3	(54.4°N, 61.7°W)				
1A	309, +28		1,0	E	290
1B	276, +45		1,0	E	290
3A	315, +01		1,0	E	290
3B	316, −05		1,0	E	290
6A	300, +24		1,0	E	290
6B	302, +30		1,0	E	290
6	**305, +21 (5.61, 13, 19)**		**28°N, 175°W**		
Site 6	(54.5°N, 61.6°W)				
2B	284, +11 (3.98)		4,0	M	672°
4A	296,0		1,0	E	250
4B	292, +17 (2.97)		3,0	M	652°
3	**291, +09 (2.97, 59, 16)**		**16°N, 166°W**		
Site 7	(54.5°N, 61.6°W)				
2A	107, −04	(107, −04)	0,1*	V	650°
3A	287, +28 (3.99)		4,0	M	670°
3B	285, +20 (6.93)		7,0	M	672°
4A	257,0 (1.87)	(259, +21)	1*,1	VM	672°
5B	293, +21 (1.91)		2,0	M	674°
5	**282, +15 (4.80, 20, 18)**		**13°N, 157°W**		
Site 8	(54.5°N, 61.6°W)				
2A	286, +05		1,0	E	674°
2B	284, +11		1,0	E	677°
3A	303, +05 (1.77)	(303, +05)	1*,1*	VM	672°
4A	294, −29 (1.96)	(294, −29)	1*,1*	VM	672°
5A	296, −03		1,0	E	680°
5B	299, −16		1,0	E	290
6	**294, −04 (5.78, 23, 14)**		**12°N, 172°W**		
Site 12	(54.5°N, 61.6°W)				
1A	253, +41		1,0	E	679°
1B	256, +46 (4.00)		4,0	M	679°
2A	245, +33		1,0	E	660°
2B	234, +37		1,0	E	679°
3A	196, +44 (5.81)		6,0	M	685°
4A	257, +20		1,0	E	290
4B	263, +19		1,0	E	679°
7	**245, +36 (6.61, 15, 16)**		**03°N, 120°W**		

TABLE 1 (*Concluded*)

Specimen	Direction $D°, I°$ (R)	VS ($D°, I°$)	w, e	Type	Trt °C/mT
		Seal diabases			
Site 2	(54.4°N, 61.7°W)				
5A	299, +22 (2.00)	(295, +26)	2*,3†	VM	60
1			**26°N, 169°W**		
Site 9	(54.4°N, 61.7°W)				
2A	295, +09 (2.00)	(115, −08)	1,1*	VM	630°
2B	297, +09		1,0	E?	630°
2	**296, +09 (2.00, 3365, 4)**		**19°N, 171°W**		
Site 11	(54.5°N, 61.3°W)				
4A	288, +14 (1.77)	(288, +14)	1*,1*	VM	630°
5A	277, +21 (1.97)		2,0	M	110
5C	285, −09 (3.83)		4§,0	M	150
3	**283, +09 (2.92, 24, 26)**		**11°N, 160°W**		
Site 13	(54.5°N, 61.6°W)				
2A	324, +20 (4.70)		5,0	M	100
5A	300, +18		1,0	E	80
5B	272, +10 (2.91)		3,0	M	80
6A	285, +10 (7.79)		8,0	M	85
6B	289, +03 (1.99)		2,0	M	80
5	**293, +13 (4.76, 17, 19)**		**19°N, 167°W**		
Site 14	(54.5°N, 61.6°W)				
1A	285, +15 (2.00)	(283, +18)	1*,1	VM	80
1B	314, +09	(314, +09)	1*,0	V	25
2A	295, +10 (1.97)		2,0	M	60
2B	318, +01 (2.96)		3,0	M	60
3A	255, +04 (5.91)		6,0	M	85
5	**294, +08 (4.61, 10, 25)**		**17°N, 169°W**		
Site 15	(54.5°N, 61.6°W)				
1B	239,0 (1.90)		2,0	M	125
2A	265, −23 (11.84)		12,0	M	120
2B	297, +24	(297, +24)	1*,0	V	110
3A	265, +22	(265, +22)	1*,0	V	75
3B	263, +23 (1.99)	(257, +24)	1*,1	VM	630°
4A	298, +34 (1.91)	(298, +34)	1*,1*	VM	115
4B	266, −03 (1.93)	(071,0)	3,1*	VM	120
5A	269, −28 (1.93)	(269, −28)	1*,3*	VM	115
5B	083, +04 (1.94)		0,2	M	115
6A	267, −18 (1.81)	(113, +28)	1,1*	VM	125
6C	275, +14 (1.93)	(275, +14)	3*,3*	VM	260
11	**269, +04 (9.96, 10, 15)**		**01°N, 150°W**		

NOTES: $D°, I°$ are declination and inclination of directions, and R the vector resultant. VS refers to the (mean) direction obtained by vector subtraction. Each specimen mean is the average of west (w) and (or) east (e) directions, which may include vector subtraction directions (*). Where both directions occur unit weight is given to each. The listed specimen direction may be a single endpoint (E) or vector subtraction (V) direction, or it may be a mean of several directions around the end-point (M) (the R refers to the number of directions averaged). If the mean is composed of several V or mixed (V with E or M) directions it is labelled VM (the R refers to 2 directions: a mean 'w' and a mean 'e'). The thermal (°C) or AF (mT) treatment (Trt) is given. Site mean statistics (in bold face) include N, the number of specimens, followed by the mean direction, the R (vector resultant), k (estimate of precision), and α_{95} (semi-angle of 95% confidence about the mean) in parentheses, and the pole obtained from the directions.

†Includes two V directions.

§One specimen direction is the mean of 3 AF directions and 1 thermal direction.

polished sections; therefore the B magnetization—if present—should predominate. Site 9 also hints at this reverse direction, and in addition hints at the presence of hematite in one thermal decay curve (not depicted). It is possible then—and this is implied by the erratic nature of the reverse direction—that there is a small A component present at site 2 which has shifted the steep reverse direction away from antiparallelism with the B direction.

A possible third and southerly directed magnetization is present at site 15. This magnetization resists fields of at least 250 mT and temperatures to

TABLE 2. Summary of *A* directions and poles

		W directions			E directions			Combined				Pole
	Mean	N	$D°,I°$	α_{95}	N	$D°,I°$	α_{95}	N	Wt	$D°,I°$ (R)	α_{95}	N°,W°(δp,δm)
						Seal volvanics						
1.	Si							5	Sp	293,+13 (4.89)	13	19,167 (6,13)
2.	Sa							21	Sa	293,+14 (19.80)	8	19,166
3.	Sp	35	294,+17	6	5	104,−07	38	36	Sp	294,+16 (33.95)	6	21,166
						Seal diabases						
4.	Si							6	Sp	289,+11 (5.88)	11	15,164 (5,11)
5.	Sa							16	Sa	283,+09 (14.84)	11	11,159
6.	Sp	26	283,+10	10	11	100,−07	16	27	Sp	283,+08 (24.64)	9	11,160
						Seal combined						
7.	Si		Group B of Roy and Fahrig (1973).					3	Sa	292,+13 (2.94)	14	18,166
8.	Si		w.r.t. horizontal					11	Sp	293,−02 (10.76)	7	12,171
9.	Si	11	291,+12	8	8	104,−13	16	11	Sp	291,+12 (10.76)	7	17,165 (4,7)
10.	Si		mean of poles					11	Sp	——— 10.84	6	17,165
11.	Si							11	Sa	290,+12 (10.73)	8	17,165
12.	Sa							37	Sa	288,+12 (34.47)	7	16,163
13.	Sp		w.r.t. horizontal					63	Sp	292,0 (58.23)	5	12,170
14.	Sp	61	289,+14	6	16	102,−07	14	63	Sp	289,+13 (58.22)	5	17,164 (3,5)

NOTES: The mean of sites (Si), samples (Sa), or specimens (Sp) is given. N is the number of sites, samples, or specimens; Wt, the unit weight given basic data; and δp, δm, the semi-axes of the ellipse of confidence about the pole for P = 0.05. Unless otherwise noted directions are given with respect to (w.r.t.) bedding. Site 12 has not been used in the averages (see text). Other symbols given in Table 1.

about 550°C, and has a large intensity (358×10^{-3} A/m; Fig. 8 (a and c)). It seems to have a reverse correlative of lower intensity (16×10^{-3} A/m) as indicated by specimen 6C in Fig. 7, which in the range from 40 to 290 mT appears to coexist with the east and west components of *A*. It is unclear whether this is the signature of an overprint at this site or the unresolved fraction of the resultant of normal and reverse *A* magnetizations.

Thermal work has indicated that the *B* direction is associated with T_{UB}'s below 600°C and the *A* with T_{UB}'s predominantly above 600°C, but ranging up to at least 690°C (highest in the volcanics). Site 15 yields an *A* direction with T_{UB}'s below 550°C. Studies of polished sections (no sections available from sites 1, 3, 6, or 14) revealed the presence of magnetite in every case. In the diabases it occurs as intergrowths with ilmenite bars and lamellae, but lacks any hematite alteration. The diabases are less altered than the volcanics (RaF), but do show selective or complete alteration of the titanomagnetite grains—possibly by hydrothermal solutions—to leave ilmenite skeletons. Hematite and (or) titanohematite was detected in all sections except those of sites 2, 9, and 15 of the diabases, the sites which yield the best evidence for the existence of the reverse *B* direction. 'Hematite' manifests itself in at least two phases: one as an alteration of the magnetite and the other as a growth occurring in fractures and as encrustations on pre-existing grains. In the volcanics, which are more highly oxidized, martitization represented by the former phase is complete.

Analysis of Directions

Two directions, *A* and *B*, as found by RaF, are clearly present. The southerly directed magnetization is well defined in only one instance and is possibly a vector resultant of normal and reverse *A* or *B* magnetizations. *A* definitely has reversals (Figs. 3 and 7) and *B* probably has reversals (Fig. 9). *A* is definitely associated with hematite, but in some instances may be associated with another magnetic mineral (e.g., in site 15). *B* and its possible reversal are probably carried by magnetite, since (1) the intensities are large, (2) the T_{UB}'s are generally < 600°C, and (3) no hematite could be detected at the reversal sites 2 and 9. The southerly directed magnetization from site 15 is associated with high AF's (> 290 mT) and T_{UB}'s between 500 and at least 550°C, and would appear not to be carried by magnetite.

The important fact to determine is the order in which these magnetizations were acquired. It was thought by RaF that (1) *A* was a secondary CRM acquired following consolidation, and (2) that *B* was probably a primary TRM. As evidence for (1) they cite generally the existence of *A* coupled with the presence of hematite; the presence of hematite being controlled by joints, irregular fractures, and

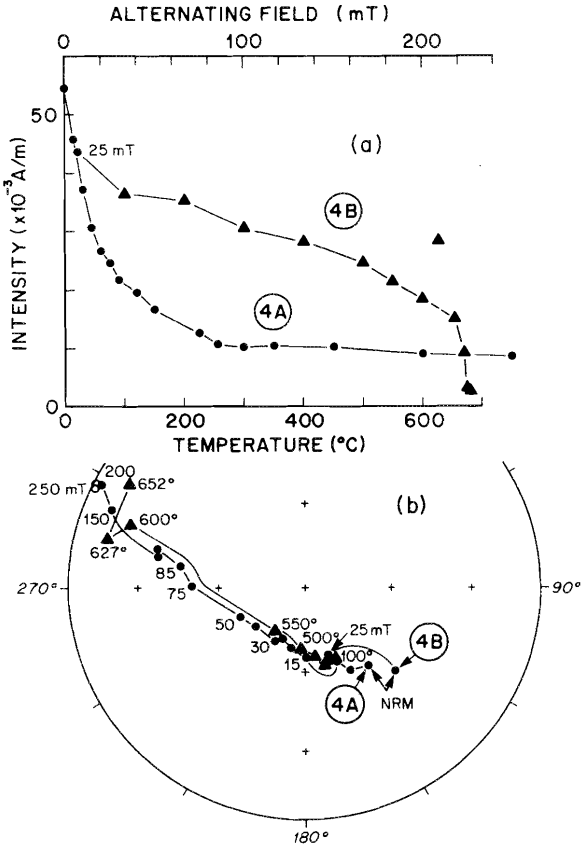


FIG. 5. Comparison of treatment by AF (4A) and two stage (4B) methods (AF followed by thermal) on separate specimens of sample 4 from site 6 to yield the *A* direction. The decay curves (a) and stereoplot (b) are shown. The intensity at 610°C is most probably an instrumental error. The directions above 652°C were inconsistent. It could not be ascertained whether these were attributable to parasitic magnetizations acquired after treatment or to unresolved dual polarity remanences.

breccia zones; and the similarity between *A* and the red-bed CRM direction. And as evidence for (2) they mention the presence of a *B*-type direction in rocks of widely different mineralogy (Croteau Group volcanics, Michael gabbro) occurring over an area of several tens of thousands of square kilometres, and the lack of evidence for post-crystallization alteration in the Michael gabbro and only evidence for hematization in the Seal-Croteau Group rocks. All of this evidence is, however, primarily evidence for the order in which magnetite and hematite formed in these rocks, and not necessarily for the order in which the present magnetizations were acquired. They note that "the shift in pole position ... between the development of the igneous (TRM) and the later hematite (CRM) ...

TABLE 3. Comparison of thermal and alternating field (AF) results

	Thermal		AF	
Site	*N*	*D*°,*I*°	*N*	*D*°,*I*°
		Seal volcanics		
1	2	294,+30	14	295,+23
3	—	———	6	305,+21
6	2	288,+14	1	296,0
7	5	282,+15	—	———
8	5	293,−02	1	291,−16
Si	4	289,+14	4	299,+07
Sp	4	288,+11	22	297,+20
		Seal diabases		
2	—	———	1	299,+22
9	2	296,+09	—	———
11	1	288,+14	2	281,+06†
13	—	———	5	293,+13
14	—	———	5	294,+08
15	1	263,+23	10	270,+02
Si	3	283,+16	5	287,+10
Sp	4	286,+14	23	283,+07
		Seal combined		
Si	7	286,+15 (6.80,11)	9	291,+09 (8.68,11)
Sp	18	288,+12 (17.11,8)	45	290,+14 (41.22,7)

NOTES: Symbols are explained in Table 1. *N* refers to specimens unless otherwise noted. Quantities in parentheses are *R* and α_{95}.
†One specimen direction is the mean of 3 AF directions and 1 thermal direction.

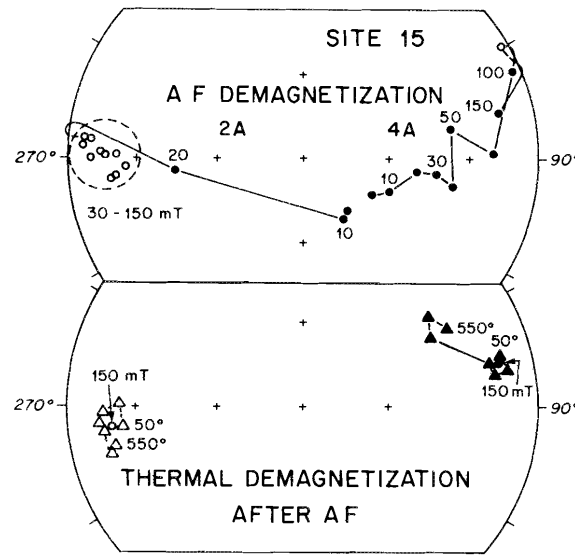


FIG. 6. Demagnetization of two specimens from site 15 directly illustrating the presence of two almost antiparallel *A* directions.

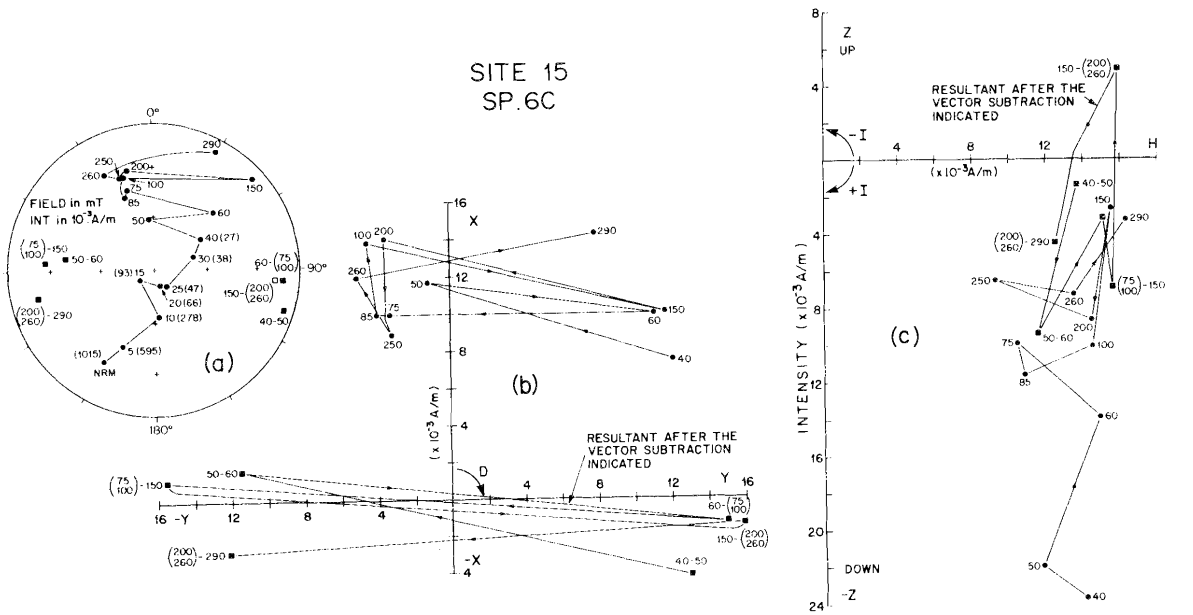


FIG. 7. Illustration of the recovery of east and west A components (squares) from a single demagnetizing specimen (dots) by the method of VS. (a) shows the stereoplot, and (b) and (c) the vector diagrams depicting the specimen and VS directions in the horizontal and vertical planes, where X and Y are the north and east components of the vector $\boldsymbol{H}$, the horizontal component ($\boldsymbol{H} = (X^2 + Y^2)^{\frac{1}{2}}$); Z, the vertical component; I, the inclination; and D, the declination. Numbers beside dots indicate first the demagnetizing field and then the intensity (if given) in parentheses. Those numbers associated with VS resultants (squares) indicate the vector subtraction performed; the numbers in parentheses indicating that all vectors between the given limits have been first averaged. The dual polarity of A is most evident in (b).

was a surprising element in this study ...[, since] the hematization of subaerial volcanic sequences is generally described as a late magmatic event" (p. 1294). If hematization is indeed a late magmatic event, then A should virtually coincide with the direction of the initial magnetization.

Our interpretation is (1) that the A magnetization residing mainly in hematite was acquired during or shortly after consolidation of the igneous rocks and (2) that the B magnetization found in magnetite is a reset magnetization of the initially formed magnetite, acquired much later, probably during cooling following uplifting of the Grenville Province.

(1) The A direction (291°, + 12°) is close to the direction (275°, + 05°) of the red-bed remanence obtained by RaF after thermal treatment. At that time, RaF interpreted the red-bed pole as being several hundred million years younger than the deposition age. However, subsequent studies of red beds have indicated that such a lapse of time between sediment deposition and the acquisition of remanence is improbable. Chemical and thermal experiments have now been performed on red-bed formations of different ages: Carboniferous (Roy and Park 1974; Roy 1977), Silurian (French and Van der Voo 1977), Cambrian (Van der Voo *et al.* 1976), and Precambrian (Henry *et al.* 1977; Morris 1977; Morris and Carmichael 1978; Roy and Robertson 1978). The results indicate that the remanence remaining after prolonged chemical leaching was acquired during and (or) shortly after deposition. Chemical leaching experiments were also performed on the Seal Group red beds, and the remarkable similarity between the chemical and thermal results (Roy and Fahrig 1973, p. 1288 and Fig. 2f) clearly indicates that the thermally determined red-bed pole relates to the initial process of magnetization. The presence of reversals in the igneous rocks indicates that the A magnetization was acquired over an extended period, part of which was at least post-cooling in the case of the volcanics if not also in the diabases. However, whether the A magnetization was obtained during cooling or within several million years of that time would make little difference in the direction. We therefore conclude that the agreement between the igneous A direction and the red-bed direction is not coincidental, and indicates that both igneous and sedimentary rock units were magnetized contemporaneously (within a few tens of Ma), shortly after emplacement and deposition of the Seal rocks.

We note that the acceptance of our interpretation

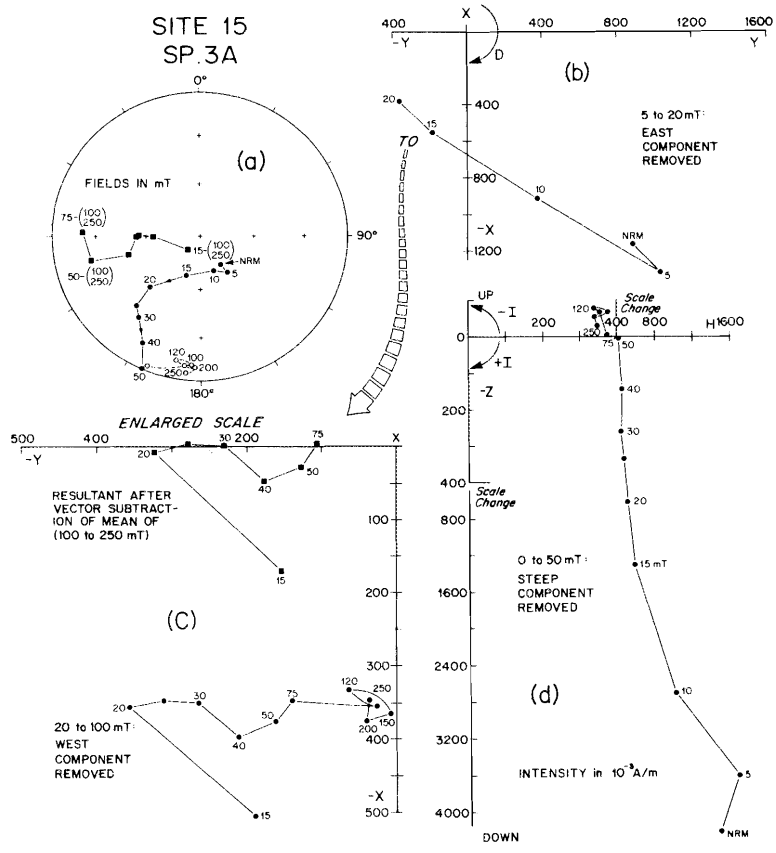


FIG. 8. Demonstration of the existence of a possible southerly direction and of the recovery by VS of a westerly *A* component. (a) depicts the stereoplot, (b) and (c) the horizontal component vector, and (d) the total vector (symbols explained in Fig. 7). Enlarged scales have been used to show more clearly the vector changes for decreasing intensity. The dots represent the data on AF treatment and the squares the resultants after the VS indicated.

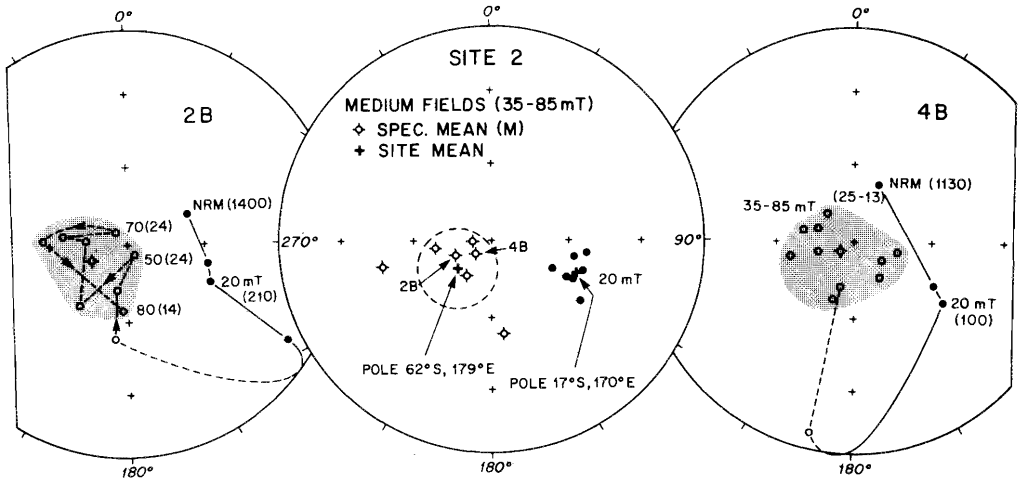


FIG. 9. A possible reversal of the *B* direction illustrated by two detailed AF demagnetization plots (specimens 2B and 4B; some fields given with intensities in parentheses) and a summary with error circle (α_{95}) (centre figure).

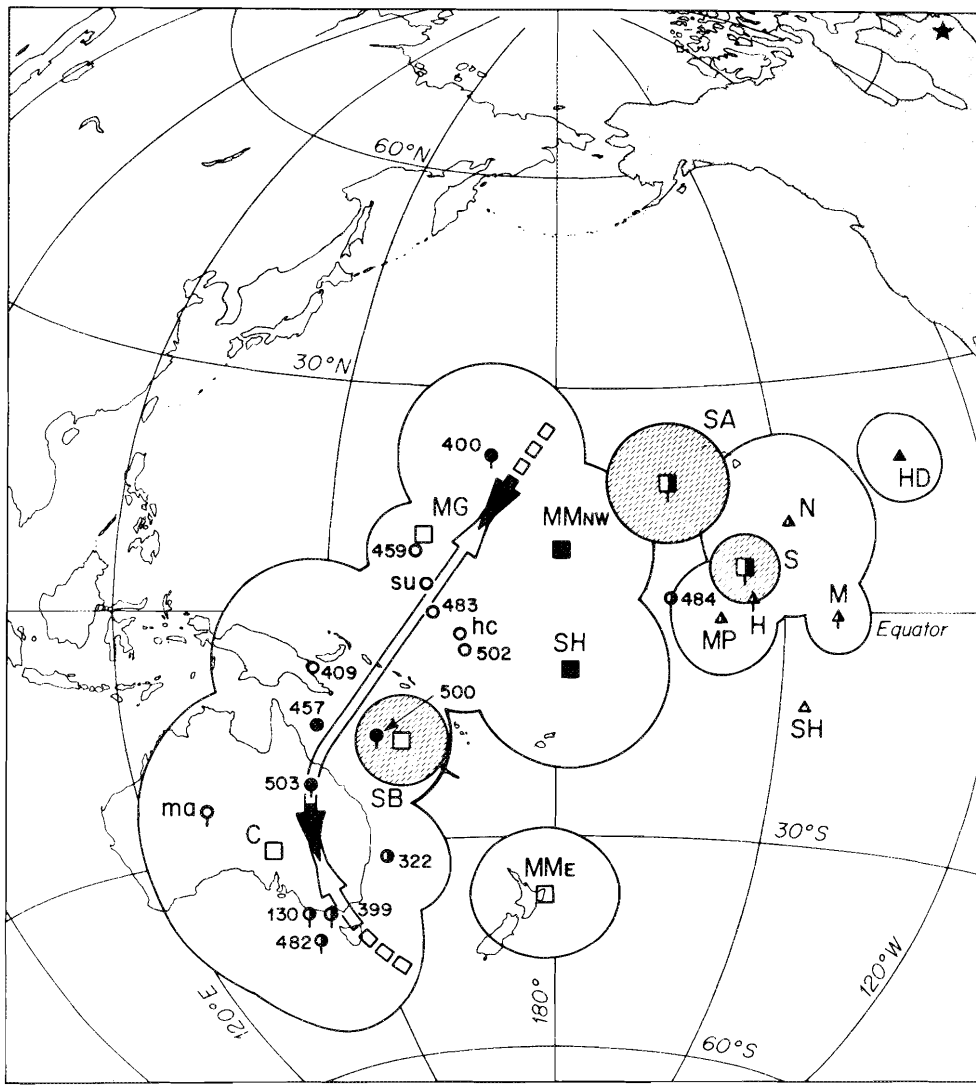


FIG. 10. Map depicting poles of Grenville (southwestern area: circles; northeastern area: squares) and Nain (triangles) Structural Provinces. Open symbols stand for magnetizations that are directed easterly and down, closed symbols for magnetizations westerly and up, and half closed symbols for magnetizations found to contain antiparallel components. 'Tailed symbols' indicate that hematite is partly or wholly responsible for the magnetization. Shaded circles are the α_{95} errors about the Seal Group poles: SB, the pole of *B* magnetization with respect to horizontal (Roy and Fahrig 1973); SA, the pole of *A* magnetization with respect to bedding (this paper); and S, the pole of red sediments with respect to bedding (this paper). Poles are identified by the Ottawa pole list number if assigned (Irving and Hastie 1975), except for those poles in the general area of the present work as identified in Fig. 1. The Grenville pole symbols not identified in Fig. 1 are: 130, Allard Lake anorthosite (Hargraves and Burt 1967); 322, Grenville metamorphic rocks, Quebec (Irving *et al.* 1972); 399, Haliburton intrusions, *A*, and 400, Haliburton intrusions, *B* (Buchan and Dunlop 1976); 409, St. Urbain anorthosite (Hargraves and Roy 1974); 457, Wilberforce pyroxenite (Palmer and Carmichael 1973); 459, Grenville Front anorthosites (Palmer and Carmichael 1973); 482, Morin complex, M1, 483, Morin complex, M2, and 484, Morin complex, M3 (Irving *et al.* 1974); 500, Whitestone anorthosite, WW, 502, Whitestone anorthosite, WY, 503, Whitestone anorthosite, WZ (Ueno *et al.* 1975); hc, Haliburton intrusions, *C* (Buchan and Dunlop 1976); ma, Magnetawan metasediments (McWilliams and Dunlop 1975); Su, Sudbury dikes at the Grenville Front (Palmer *et al.* 1977). The Mealy Mountain northwest (MM_{NW}) and east (MM_E) component poles are shown. The Shabogamo gabbro poles SH have been separated into two groups according to whether the sites were obtained at the Grenville Front (13 sites) (square) or north (3 sites) of the Front in the Nain Province (triangle). The given arrow is merely to indicate the trend of the Grenville polar curve as suggested by previous workers, whether ascending, descending, or both. The star indicates the sampling area.

TABLE 4. *B* directions and poles

	Direction	Pole
	$D°,I°(N,R,k,\alpha_{95})$	S°,E°($\delta p,\delta m$)
Seal Group	125,+61(14,13.70,43,6)	17,158(7,9)
Croteau Group	142,+75(5,4.85,26,15)	30,138(26,28)

NOTES: Symbols explained in preceding Tables. Directions with respect to horizontal.

greatly weakens the rotation hypothesis of RaF. The hypothesis was based on the fact that some of the non-uniformly magnetized red beds yielded a direction intermediate (~ 330°) between the dual polarity mean direction (275°, + 05°). The interpretation then was that "part of the non-uniformly directed magnetization ... was acquired during the field reversal and/or rotation of the area" (p. 1289). Figure 4 of RaF, however, shows great similarity to Fig. 4 of Roy (1977) where it was shown by slicing specimens that the intermediate directions were attributable to the occurrence of a field reversal during the magnetization process. This has been substantiated in other studies, some of which are quoted above. It then appears that, of the two alternatives offered by RaF, the former is to be preferred.

(2) If the *A* magnetization is the initial remanence, the *B* magnetization whose direction differs by more than 70° (irrespective of sign) from that of *A* is necessarily a secondary or reset magnetization. Its age cannot be determined directly, but can be inferred from other evidence. The *B* pole falls on the Grenville polar curve (Fig. 10), which is composed of poles derived from magnetizations commonly attributed to resetting during cooling following Grenville metamorphism (Buchan and Dunlop 1976). It is therefore possible that magnetite, having lower T_{UB}'s than hematite, may have been the only carrier to be subjected to resetting during the Grenville orogeny. This possibility has also been suggested by Ueno and Irving (1976) with regard to the rocks from the Chibougamau area and others near the Grenville Front. Furthermore, although the Croteau Group predates the Seal Group by 200 Ma (W. R. A. Baragar *in* Wanless and Loveridge 1978), the *B* directions of the two units are in agreement within error (Table 4). The agreement may be owing to a coincidental juxtaposition of the related polar loops, to a stationary apparent pole during the interval, or to a later concurrent remagnetization of both rock units. The fact that both poles fall on the Grenville polar curve supports the third alternative, indicating the *B* magnetizations of the two rock units were produced during the Grenville orogeny.

Conclusions

The poles obtained from the Nain and Grenville Structural Provinces are shown in Fig. 10. The Nain poles (triangles) form a cluster near the equator at about 205°E, and both the igneous *A* (SA) and red-bed (S) Seal Group poles belong to that group. The Grenville poles (circles and squares) lie mainly in a belt stretching north-northwest from south of Australia (140°E, 40°S) to the North Pacific Ocean (170°E, 20°N) and the *B* poles of both the Seal (SB) and Croteau (C) Group igneous rocks, plotted with respect to horizontal rather than bedding as in RaF, lie within this group. The Nain rocks date generally between 1450 and 1350 Ma. The ages of Grenville poles are not accurately known, but are believed to be within the 1200 to 950 Ma interval. Detailed AF and thermal experiments and re-interpretation of the chemical experiments of RaF indicate that both the igneous *A* and the red bed hematite magnetizations were acquired shortly after consolidation of the rocks. Both the SA and S poles are therefore associated with the 1450 to 1350 Ma interval. The magnetite *B* magnetization was acquired later, and because the pole (SB) falls amongst the Grenville poles, it is associated with the 1200 to 950 Ma interval. This does not disprove the model presented by RaF, but offers an alternative which does not require a tectonic rotation based on red-bed results that can be explained otherwise.

It is interesting that the Seal Group igneous (SA) and red-bed (S) poles agree relatively well with the pole of the probably uncomformably underlying Harp Lake Complex (H), even though the latter is probably about 100 Ma older (Irving *et al.* 1977). This apparent anomaly, as well as the separation between the '1400 Ma' poles and the Grenville poles to the west, illustrates the gaps in the polar record. Both groups may represent parts of several polar loops (e.g., Irving *et al.* 1977; Murthy 1978), but before these can be convincingly defined there must be a significant improvement in radiometric age dating, and a more precise way of relating dates obtained by various methods to the various magnetizations in the rock.

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Addendum

The behaviour of resultant magnetizations composed of normal and reverse components having different T_{UB} or rcf spectra is investigated here by means of a theoretical model. Knowing the thermal demagnetization curves of each component (Fig. 11a), and their directions at each temperature (Fig. 11b: A, B, C, D, E, F), the resultant directions (Fig. 11b: X, Y, Z) (Table 5) and thermal curves (Fig. 11c, d, e) have been generated. In the data used, different unblocking temperatures record a slightly different field direction (numbered points correspond in each part of Fig. 11). The westerly cluster of directions for convenience represents the normal component from three theoretical specimens (1, 2, and 3). Each of these normal components (A, E, and F) is combined with a reverse component: E with C, which is almost antiparallel, and A and F with B and D, which are 10 to 20° away from antiparallelism. Intensity data are read from the artificial intensity curves (Fig. 11a). As an example, the resultant direction and intensity of specimen 1 at 660°C (point 3) is represented in Figs. 11b and c by the point X3. The resultant represents the vector sum of the normal vector A3 and the reverse vector D3 with intensities given in Fig. 11a and directions in Fig. 11b. The behaviour of the synthesized data of the normal and reverse components clearly models the type of behaviour displayed by the specimens of the present and other studies.

With the resultant X, Y, and Z data, representing data that one might ordinarily obtain from the thermal demagnetization of several specimens, it is attempted as an exercise to arrive at the initial normal and reverse components by using vector subtraction (the justification for this is given below). To cover all vector subtraction possibilities the vector at each temperature has been subtracted from all the individual vectors at lower temperatures and the resulting directions plotted in Fig. 12. The question is: How are we to know which VS

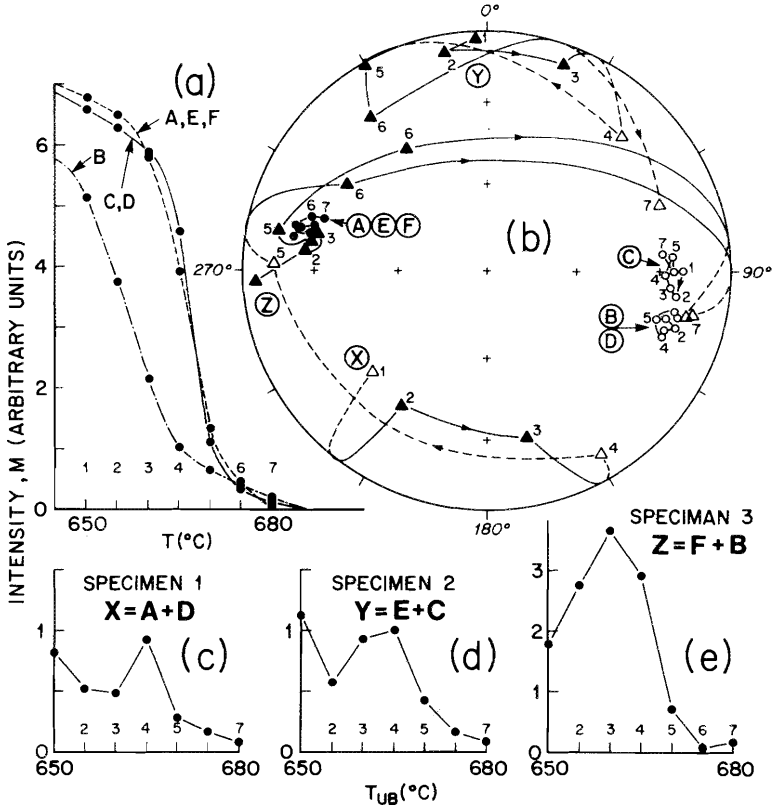


FIG. 11. A model to illustrate the behaviour of a specimen containing antiparallel magnetizations. (a) depicts the thermal decay curves between 650 and 680°C of the normal (A, E, F) and reverse (B, C, D) components of three specimens (1, 2, 3). (b) shows the directions of all components and their vector resultants (X, Y, Z) at each temperature (numbers correlate with temperature scale). (c), (d), and (e) are the thermal decay curves showing the intensities of the vector resultants **X**, **Y**, and **Z**. The resultants **X**, **Y**, and **Z** are what one would actually observe during the thermal demagnetization of the specimens.

TABLE 5. Resolution of magnetizations having normal and reversed components

BT (°C)	First vector Sym	First vector M,D°,I°	Second vector Sym	Second vector $M,D^\circ I^\circ$	Resultant vector Sym	Resultant vector M,D°,I°
SPECIMEN 1						
			Synthesis (Vector addition)			
650	A1	6.77,280,+20	D1	6.60,105,−25	X1	0.81,228,−36
655	A2	6.50,283,+23	D2	6.28,107,−20	X2	0.58,206,+43
660	A3	5.80,285,+27	D3	5.89,109,−24	X3	0.49,167,+29
665	A4	3.92,282,+26	D4	4.60,111,−24	X4	0.92,148,−10
670	A5	1.32,283,+20	D5	1.10,106,−28	X5	0.28,272,−13
675	A6	0.45,287,+25	D6	0.33,102,−22	X6	0.13,302,+32
680	A7	0.05,288,+30	D7	0.13,104,−20	X7	0.08,102,−14
		284,+24		*106,−23*		
Mean		**285,+24**				
			Analysis (Vector subtraction: VS)			
	X1		X4			1.07,289,−17
	X2		X3			0.33,277,+29
	X2		X4			0.95,300,+36
	X3		X4			0.65,312,+37
	X4		X5			1.09,136,−05
	X5		X7			0.35,274,−07
	X6		X7			0.21,295,+25
	X7		—			0.08,102,−14
Mean (N, R, k, α_{95})		**293,+16 (8, 7.35, 11, 18)**				
SPECIMEN 2						
			Synthesis (Vector addition)			
650	E1	6.77,280,+20	C1	6.60,090,−20	Y1	1.11,357,+03
655	E2	6.50,283,+22	C2	6.28,098,−22	Y2	0.56,349,+08
660	E3	5.80,285,+27	C3	5.89,095,−25	Y3	0.94,020,+09
665	E4	3.92,282,+26	C4	4.60,091,−27	Y4	1.00,045,−22
670	E5	1.32,283,+20	C5	1.10,085,−24	Y5	0.42,330,+01
675	E6	0.45,287,+25	C6	0.33,090,−24	Y6	0.16,323,+21
680	E7	0.05,288,+30	C7	0.13,084,−28	Y7	0.09,071,−25
		284,+24		*090,−24*		
Mean		**277,+24**				
			Analysis (VS)			
	Y1		Y3			0.45,302,−11
	Y1		Y4			0.95,302,+27
	Y2		Y3			0.54,232,−07
	Y2		Y4			0.89,262,+30
	Y4		Y5			0.99,071,−23
	Y4		Y6			1.02,054,−25
	Y5		Y7			0.45,320,+06
	Y6		Y7			0.21,299,+27
Mean (N, R, k, α_{95})		**276,+18 (8, 6.60, 5, 28)**				
SPECIMEN 3						
			Synthesis (Vector addition)			
650	F1	6.77,280,+20	B1	5.12,105,−25	Z1	1.79,267,+05
655	F2	6.50,283,+22	B2	3.75,107,−20	Z2	2.77,277,+25
660	F3	5.80,285,+27	B3	2.15,109,−24	Z3	3.66,283,+29
665	F4	3.92,282,+26	B4	1.02,111,−24	Z4	2.91,279,+27
670	F5	1.32,283,+20	B5	0.63,106,−28	Z5	0.70,281,+13
675	F6	0.45,287,+25	B6	0.40,102,−22	Z6	0.06,327,+39
680	F7	0.05,288,+30	B7	0.20,104,−20	Z7	0.15,103,−17
		284,+24		*106,−23*		
Mean		**285,+24**				

TABLE 5 (*Concluded*)

BT (°C)	First vector Sym	First vector M,D°,I°	Second vector Sym	Second vector M,D°,I°	Resultant vector Sym	Resultant vector M,D°,I°
			Analysis (VS)			
	Z1		Z3			2.25,121,−46
	Z2		Z3			0.96,123,−39
	Z3		Z5			2.99,284,+33
	Z3		Z6			3.61,282,+29
	Z3		Z7			3.81,283,+29
	Z4		Z5			2.24,278,+31
	Z4		Z6			2.87,278,+27
	Z4		Z7			3.06,279,+27
	Z5		Z7			0.85,281,+14
	Z6		Z7			0.20,293,+24
	Z7		—			0.15,103,−17
Mean (N, R, k, α_{95})			**285,+29 (11, 10.79, 48, 7)**			
True mean			**282,+24 (3, 3.00, 365, 7)**			
V.S. mean			**285,+21 (3, 2.97, 58, 16)**			

NOTES: Examples of the synthesis and analysis of normal and reverse magnetic components in single specimens are given. The data are artificial, so that resolved components can be compared to true values. Intensities of separate components have been read from the artificial unblocking temperature (T_{UB}) curves of Fig. 11a. Directions of individual components and resultants are plotted in Fig. 11b, where vector symbols (Sym) correspond to those of this table. Data given under 'Analysis' are shown shaded in Fig. 12. Specimen means (in bold face) are given after each set of data, and the final means (bold face) with their statistics are at the end. Symbols are explained in preceding Tables. The means under the syntheses have been calculated by first averaging the east and west components separately (italicized means) and then meaning the two together.

direction will give us the best approximation to the true direction? For specimens 1 and 2 it is apparent that there is a trend of VS directions between the normal and reverse components; but if we choose the point at each end of the trend we would be in error by 30 to 50°. For specimen 3 there is a cluster of directions in the normal and reverse directions, and the estimates would be much better. It is apparent from the VS results plotted in Fig. 12 that those directions which most closely approximate the true direction are mainly those (indicated by shading) obtained by using the vectors that produce relative maximum or minimum values on the demagnetization curves or which plot as extrema directions, that is, points at which the migration of the vector under demagnetization changes by about 180°. Thus one would subtract from one another vectors represented by adjacent extrema of intensity or direction.

This VS procedure has been carried out for all three specimens by only regarding the synthesized data (Fig. 11b and c). The individual VS directions and means are given in Table 5. The VS mean (285°, +21°; α_{95} = 16°) agrees within error with the true mean (282°, +24°).

Why should VS of adjacent extrema of direction and intensity yield the best estimates of the true direction? Let us consider two successive vectors $X_n - X_m$ (e.g., resultant vectors of specimen 1). Each is composed of a normal vector, A, and a reverse vector, D, such that

$$X_n - X_m = (A_n + D_n) - (A_m + D_m)$$

or

$$X_n - X_m = (A_n - A_m) + (D_n - D_m)$$

If we examine the relation $(A_n - A_m)/(D_n - D_m)$, it is apparent that as this quantity becomes larger or smaller away from unity, the approximation of $X_n - X_m$ to A or D improves. This occurs because there is less or more of a change in one of the components between successive steps. When the relative rate of change of decay between the vectors reverses, the result is an aberration or relative extreme in the intensity decay curve or stereoplot. In the extreme case, when only one vector changes between demagnetization steps, the other is found exactly (Buchan and Dunlop 1976). The degree of divergence of the above ratio from unity is indicated by the numbers in parentheses in Fig. 12. It is apparent in general that as the VS direction approaches the true direction the numbers become larger.

The more general question to ask is: Why should any sensible direction at all be forthcoming from the VS of two resultant vectors representing components which overlap substantially? The answer lies in the fact that we are dealing with resultants of

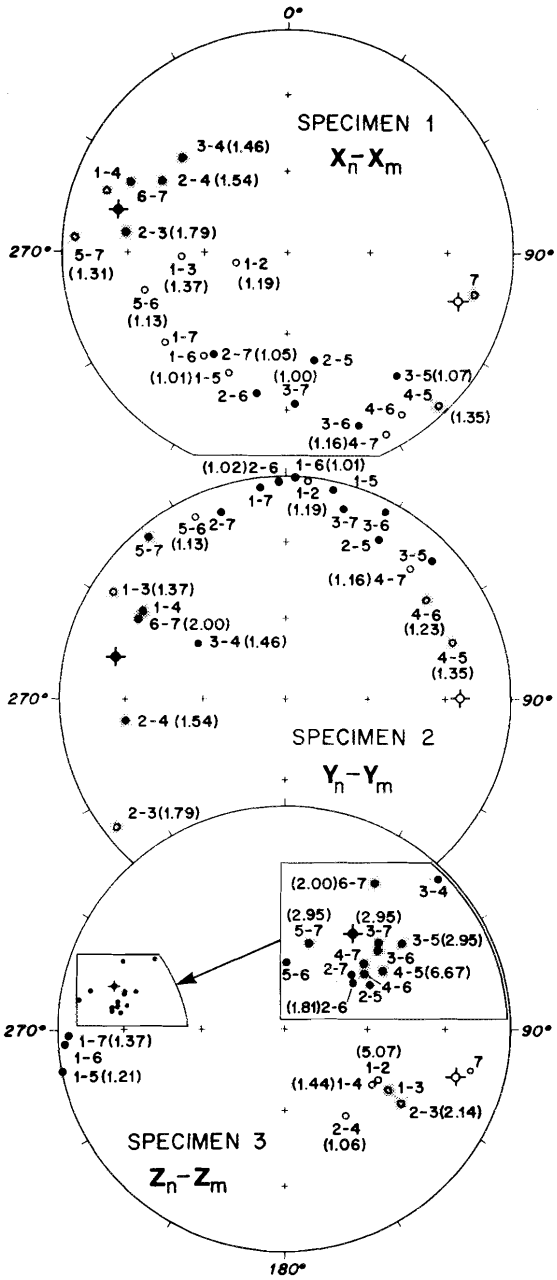


FIG. 12. The VS directions obtained by performing the vector subtractions indicated beside each point. All possible VS directions are plotted. The numbers correlate with those in Fig. 11. Shaded points are those VS directions considered on the basis of the observed data (resultants in Fig. 11b, c, d, and e) to approximate the true directions (crossed symbols) (see Table 5). Numbers in parentheses indicate values of the statistic $(A_n - A_m)/(D_n - D_m)$, or its correlatives for specimens 2 and 3, such that numerator and denominator may be interchanged in order to yield numbers greater than one (see text).

two *nearly opposed* vectors. In this case it is clear that small differential changes between components will result in large changes in the direction of the resultant vector. And the largest differences between resultant vectors themselves will occur at the extrema values, resulting in the best approximation to the true direction. The major pitfall in the VS method when dealing with nearly opposed magnetizations is that any small noise components, being comparable in size to the magnitude of vector resultants of the stable components, can conceivably result in large errors. However, if the noise is random it should be eliminated as a bias factor when many VS directions are averaged.

This exercise shows that decomposition of dual polarity remanences is not easily accomplished. Without careful inspection of the data, much information could be missed; for example in specimen 3, the east component is revealed directly only at 680°C (Fig. 11b). The vector analysis, however, clearly shows a clustering of easterly directions which involve the entire T_{UB} spectrum (Fig. 12, specimen 3). What is even worse is that data could be wrongly interpreted. The Y directions of specimen 2 could possibly be mistaken for a northerly-directed remanence (Fig. 11b). It might even be possible to find support for this assumption through the careless use of VS. Note, for example, the high concentration of directions obtained by subtracting the following: 1–2, 1–5, 1–6, 1–7, 2–6, 2–7, 3–6, and 3–7 (Fig. 12, specimen 2). It then appears that interpretation of two distinct remanences as opposed to dual polarity remanences must be approached with caution.